Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese

Department of Electrical, Electronic and Computer Engineering

Semestertoets 2
VRAESTEL
Kopiereg voorbehou

Semester Test 2
QUESTION PAPER
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Module EBN122
Elektrisiteit en Elektronika
October 2017

Module EBN122
Electricity and Electronics
October 2017

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*			
Maksimum punte: *Maximum marks:*	46	Volpunte: *Full marks:*	45
Duur van vraestel: *Duration of paper:*	90 min + 10 min om OKH vorm te voltooi *90 min + 10 min to complete OCR form*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*
Totale aantal bladsye (hierdie blad ingesluit): *Total number of pages (including this page):*		18 + OCR	

BELANGRIK- ***IMPORTANT***

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
   The examination regulations of the University of Pretoria apply.
2. **Alle vrae moet op die op die Optiese Karakter Herkenning (OKH) vorm ingevul word.**
   All questions must be answered on the Optical Character Recognition (OCR) form.
3. Neem noukeurig kennis van die instruksies op die OKH vorm.
   Take careful consideration of the instructions on the OCR form.
4. Antwoorde moet eenhede insluit!
   Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
   Final answers must be written as real numbers, and not as fractions.

Dosent(e): *Lecturer(s):*	1. D le Roux
	2. D Ramotsoela
	3. X Ye

INSTRUCTIONS

Do step 0 at the start of the session.
Do step 1 in the first 90 minutes of the test
Do step 2 within the following 10 minutes.

STEP 0 (-5 min to 0 min): Complete the front page of the OCR form

STEP 1 (0 min to 90 min): Do your own calculations on the QUESTION PAPER. The question paper **will not** be taken in at the end of the test. Write down the answers for each question on the PRACTICE ANSWER SHEET. Be sure to include the units. The PRACTICE ANSWER SHEET **will not** be taken in at the end of the test.

STEP 2 (90 min to 100 min): Carefully copy the answers from the PRACTICE ANSWER SHEET to the OCR form.
- The Assessment ID is: **2017EBN122S2**
- Include the correct unit for each answer. Apply appropriate prefixes.
- Use a mechanical pencil to complete the form.
- Each cell should only have one character.
- Characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops.

Examples

01	–	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2						k	Ω							
04	-	1																

INSTRUKSIES

Doen stap 0 voor die begin van die sessie.
Doen stap 1 in die eerste 90 minute van die toets.
Doen stap 2 binne die daaropvolgende 10 minute.

STAP 0 (-5 min to 0 min): Voltooi die voorblad van die OKH vorm

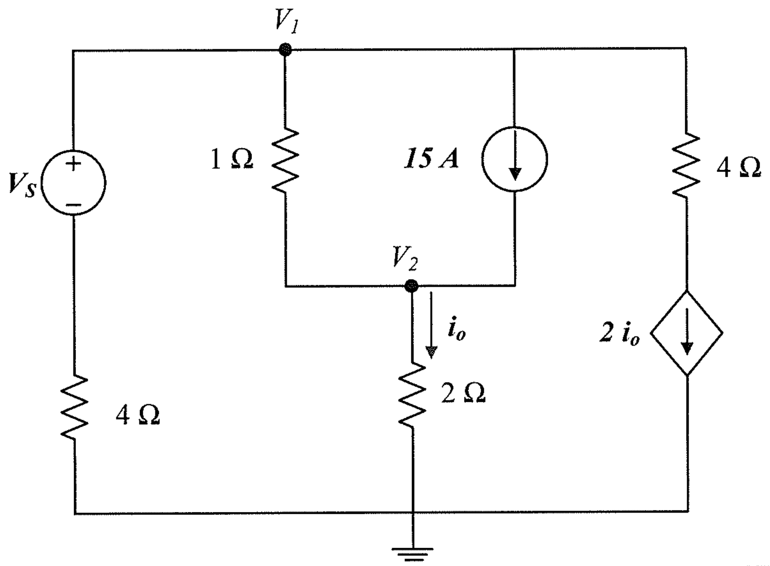
STAP 1 (0 min to 90 min): Doen jou eie berekeninge op die VRAESTEL. Die VRAESTEL **word nie** aan die einde van die toets ingeneem nie. Skryf die antwoorde vir elke vraag op die OEFEN ANTWOORDBLAD. Maak seker om eenhede in te sluit. Die OEFEN ANTWOORDBLAD **word nie** ingeneem nie.

STAP 2 (95 min to 100 min): Dra die antwoorde vanaf die OEFEN ANTWOORDBLAD oor na die OKH vorm.
- Die "Assessment ID" is **2017EBN122S2**
- Sluit die korrekte eenheid vir alle antwoorde in. Maak gebruik van die toepaslike voorvoegsel.
- Gebruik 'n meganiese potlood om antwoorde in te vul.
- Elke blokkie mag slegs een karakter bevat.
- Karakters mag nie aan die grense van die blokkies raak of oor dit steek nie.
- **Moenie kommas gebruik nie!** Maak eerder gebruik van punte.

Voorbeelde

01	–	1	.	4	5					m	A							
02	3	4	0							V								
03	1	5	.	2						k	Ω							
04	-	1																

Question 01-02 [4]		Vraag 01-02 [4]	
01. Find the exclusive contribution of the 15 A current source to i_0. (Hint: Use nodal analysis at V_1 and V_2.) 02. The exclusive contribution of the voltage source V_S to i_0 is: $i_{0(V_S)} = -4$ A. Calculate V_S.		01. Bepaal die eksklusiewe bydra van die 15 A stroombron tot i_0. (Wenk: Gebruik node-analise by V_1 en V_2.) 02. Die ekslusiewe bydra van die spanningsbron V_S tot i_0 is $i_{0(V_S)} = -4$ A. Bereken V_S.	
01	$i_{o(15A)}$	**01**	$i_{o(15A)}$
02	V_S	**02**	V_S



01: Let $V_S = 0$, and use nodal analysis

V_1: $\frac{V_1}{4} + V_1 - V_2 + 15 + 2 \cdot \frac{V_2}{2} = 0$

V_2: $V_1 - V_2 + 15 = \frac{V_2}{2}$

$\Rightarrow V_1 = -12, \quad V_2 = 2, \quad \underline{i_0 = 1\text{ A}}$

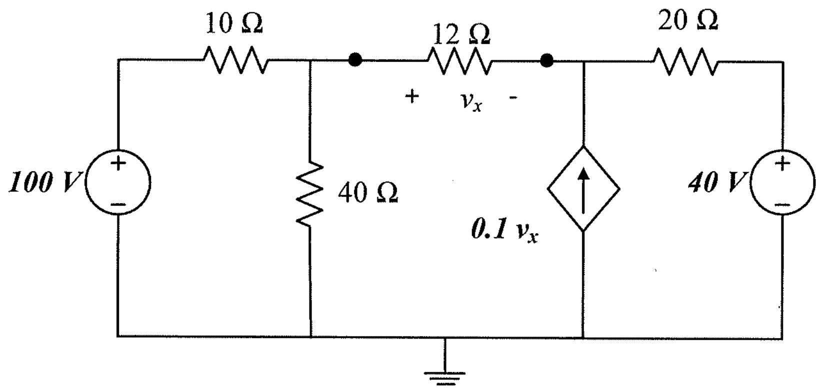
02: Let 15A current source open circuit, then

$V_1 = -4 \times 3 = -12$V.

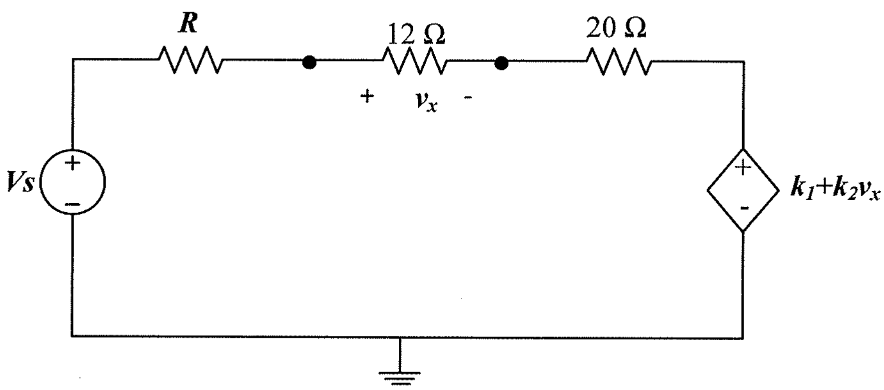
At V_1: $\frac{V_1 - V_S}{4} + (-4) + (-8) = 0$

$\Rightarrow \underline{V_S = -60\text{ V}}$

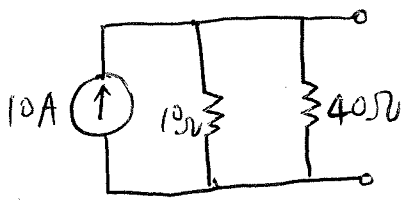
Question 03-06 [4]		Vraag 03-06 [4]	
Circuit (a) can be transformed into Circuit (b). Calculate k_1, k_2, R, and V_S.		Kringbaan (a) kan in Kringbaan (b) getransformeer word. Bepaal k_1, k_2, R, en V_S.	
03	k_1	**03**	k_1
04	k_2	**04**	k_2
05	R	**05**	R
06	V_S	**06**	V_S



↑ **Circuit (a)** ↑



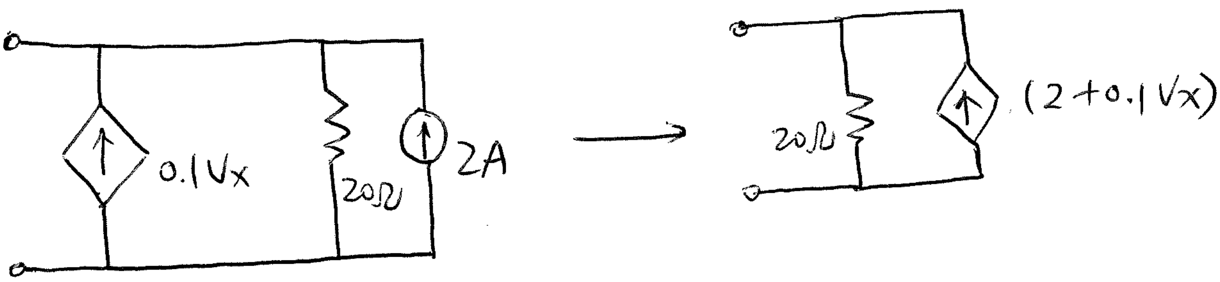
↑ **Circuit (b)** ↑



,

$R = 10 \| 40 = 8\,\Omega$

$V_S = 10A \times 8\,\Omega = 80\,V$

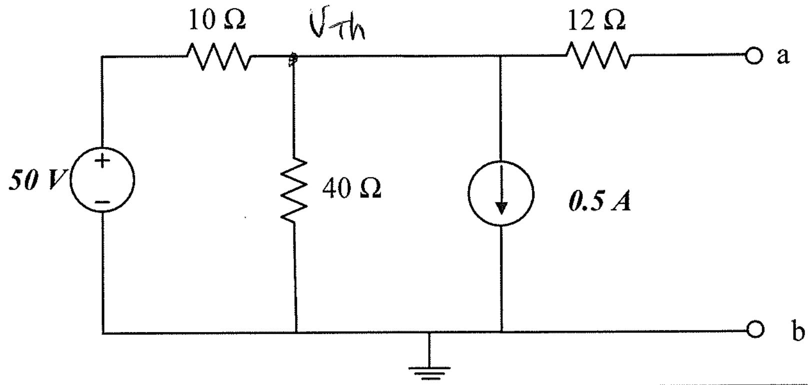


$V = 20(2+0.1V_x)$
$= 40 + 2V_x$

$\Rightarrow k_1 = 40$

$k_2 = 2$

Question 07-08 [4]		Vraag 07-08 [4]	
Find the Thevenin equivalent V_{Th} and R_{Th} with respect to terminals a-b.		Bepaal die Thevenin ekwivalent V_{Th} en R_{Th} met verwysing tot terminale a-b.	
07	V_{Th}	07	V_{Th}
08	R_{Th}	08	R_{Th}



$$R_{Th} = 12 + 10 // 40 = 20\Omega$$

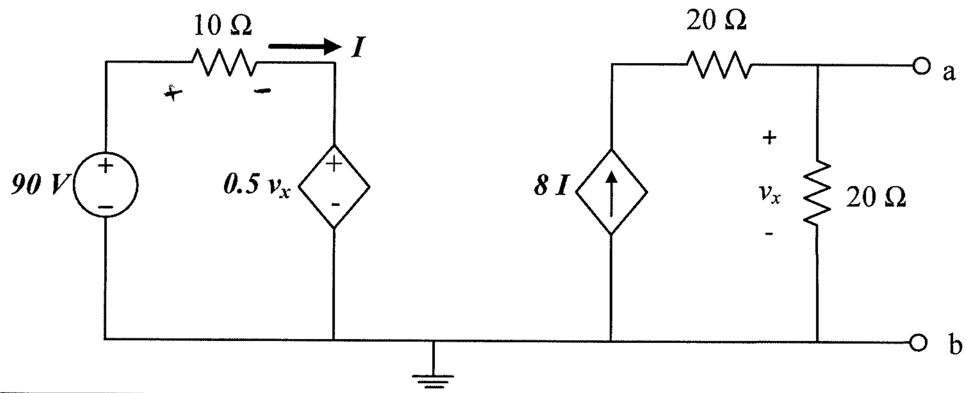
$$\frac{V_{Th} - 50}{10} + \frac{V_{Th}}{40} + 0.5 = 0$$

$$\Rightarrow 4V_{Th} - 200 + V_{Th} + 20 = 0$$

$$\Rightarrow 5V_{Th} = 180$$

$$\Rightarrow \underline{V_{Th} = 36\ V}$$

Question 09-10 [4]		Vraag 09-10 [4]	
Find the Norton equivalent I_N and R_N with respect to terminals a-b.		Bepaal die Norton ekwivalent I_N en R_N met verwysing tot terminale a-b.	
09	I_N	09	I_N
10	R_N	10	R_N



$I_N = 8I$, when a-b is shorted, $V_x = 0$

$I = \frac{90}{10} = 9\,A \Rightarrow I_N = 72\,A$.

Get V_{oc} for R_N, $V_{oc} = V_x$,

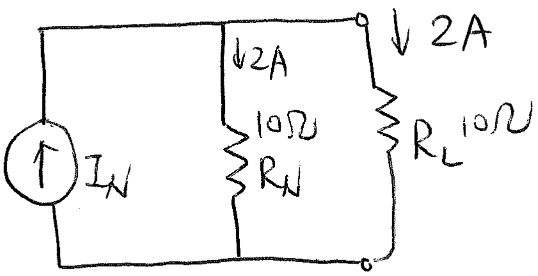
$V_x = 20 \times 8I = 160 \cdot \frac{90 - 0.5V_x}{10}$

$V_x = 16(90 - 0.5V_x) = 16 \times 90 - 8V_x$

$\Rightarrow 9V_x = 16 \times 90 \Rightarrow V_x = 160\,V.$

$R_N = \frac{20}{9} = 2.2222\,\Omega$

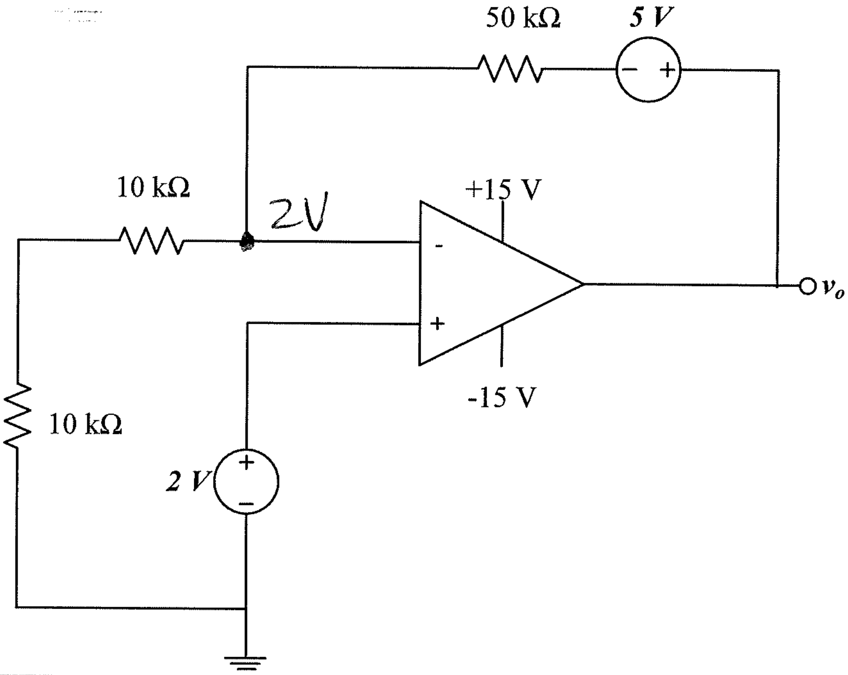
Question 11-12 [2]		Vraag 11-12 [2]	
A Norton equivalent circuit is connected to a load resistor R_L. Maximum power P_{max} is delivered to the load R_L. If the current through the load R_L is 2 A and $R_N = 10\ \Omega$, find the Thevenin equivalent voltage V_{Th} and the value of P_{max}.		'n Las weerstand R_L word aan 'n Norton ekwivalente kringbaan gekoppel. Maksimum drywing P_{max} word aan die las R_L voorsien. Indien die stroom deur die las R_L gelyk is aan 2 A, en $R_N = 10\ \Omega$, bepaal die Thevenin ekwivalente spanning V_{Th} en die waarde van P_{max}.	
11	V_{Th}	**11**	V_{Th}
12	P_{max}	**12**	P_{max}



$I_N = 4A$, $V_{Th} = 4 \times 10 = 40V$

$$P_{max} = \frac{V_{Th}^2}{4R_{Th}} = \frac{40^2}{4 \times 10} = 40W$$

Question 13 [2]		Vraag 13 [2]	
13	Find v_o	13	Bepaal v_o

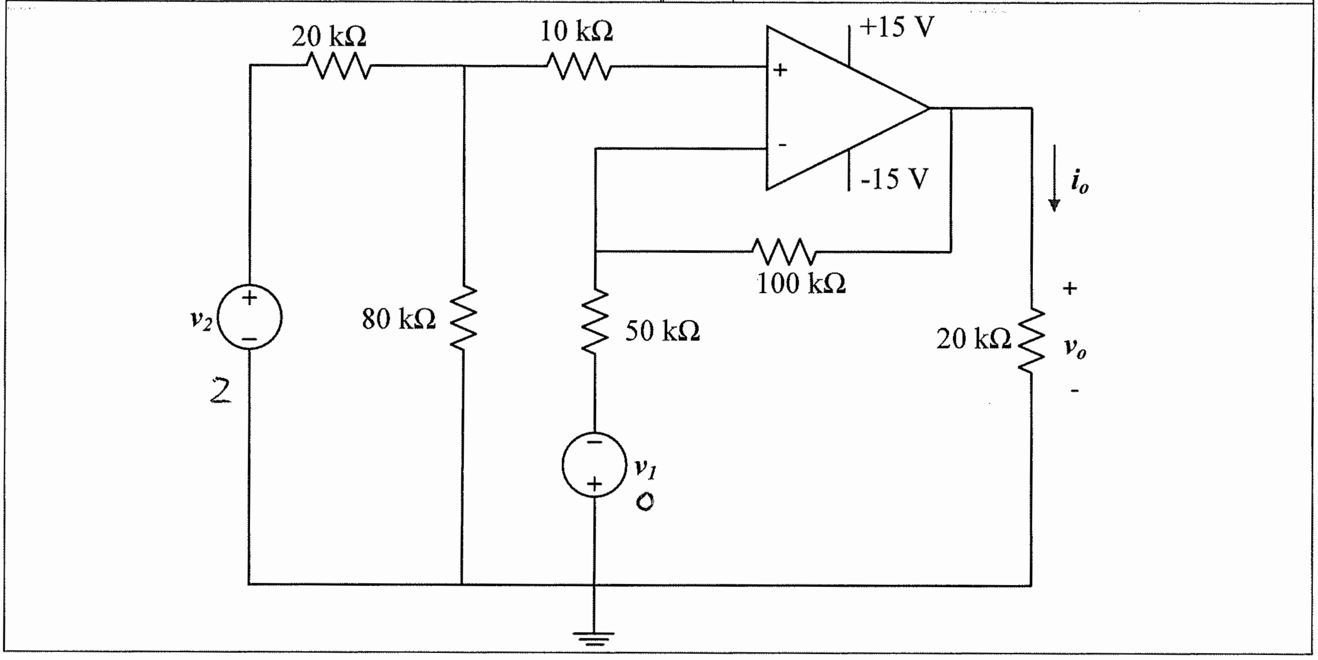


$$\frac{2}{20} = \frac{V_o - 5 - 2}{50}$$

$$100 = 20V_o - 140$$

$$\Rightarrow V_o = 12V$$

Question 14-15 [4]		Vraag 14-15 [4]	
14	Find v_o if $v_1 = 0$ V and $v_2 = 2$ V.	**14**	Bepaal v_o indien $v_1 = 0$ V en $v_2 = 2$ V.
15	Find i_o if $v_1 = 2$ V and $v_2 = 0$ V.	**15**	Bepaal i_o indien $v_1 = 2$ V en $v_2 = 0$ V.



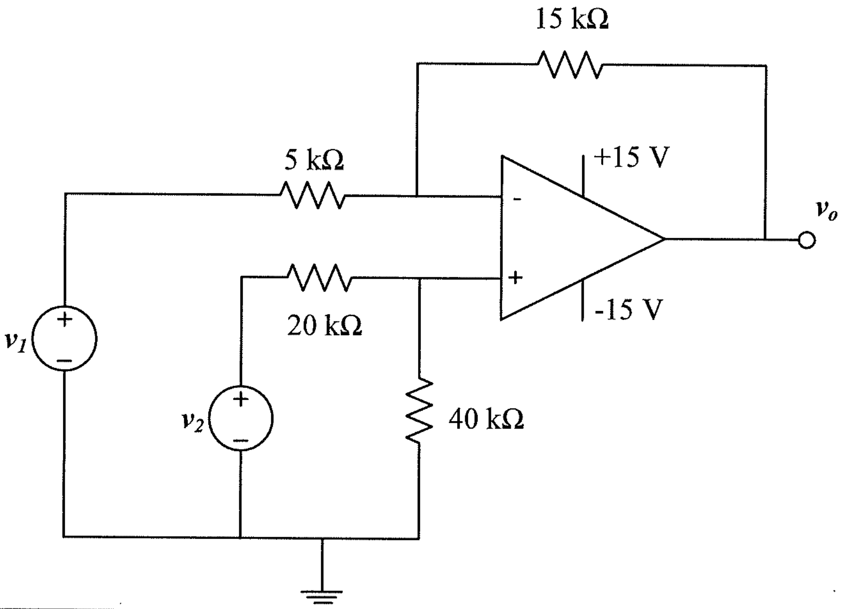
14: $V^+ = 2 \times \frac{80}{20+80} = 1.6$ V.

$V_o = V^+\left(1 + \frac{100}{50}\right) = \cancel{2.4V}$ **4.8 V**

15: $V_o = -(-2) \cdot \frac{100}{50} = 4V$

$i_o = \frac{4V}{20k\Omega} = 0.2$ mA

Question 16-17 [4]		**Vraag 16-17 [4]**	
01. Find v_o if $v_1 = 3$ V and $v_2 = 6$ V. 02. If $v_1 = 3$ V, then find the maximum value of v_2 to avoid positive saturation of the amplifier.			
16	v_o	**16**	v_o
17	v_2^{max}	**17**	v_2^{max}



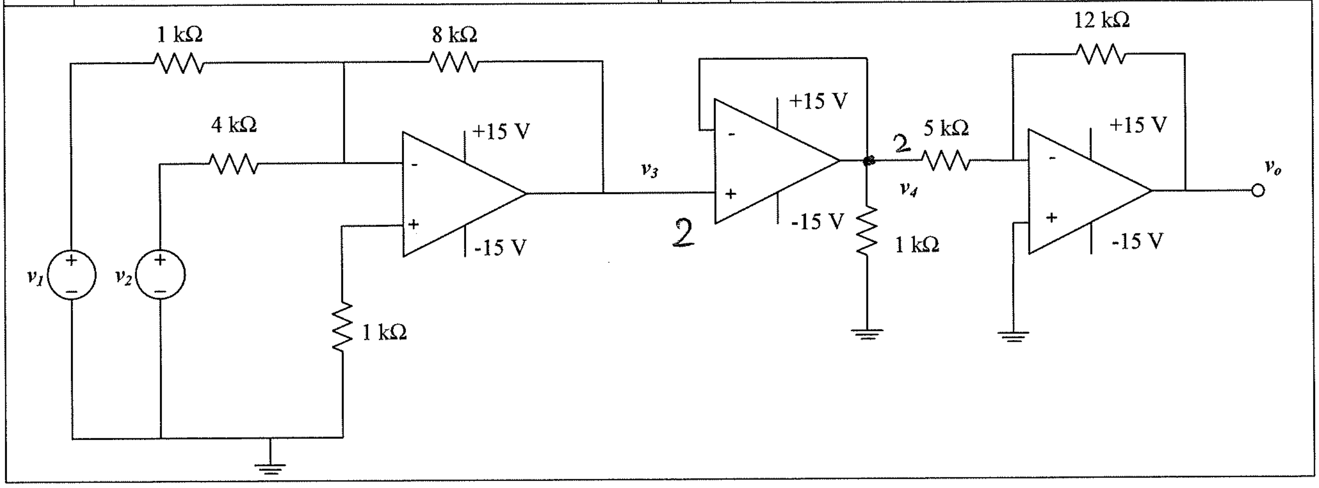
$$V_o = \frac{15\left(1+\frac{5}{15}\right)}{5\left(1+\frac{20}{40}\right)} \times 6 - \frac{15}{5} \times 3 = 7V$$

$$15 = \frac{4}{1.5} \times V_2 - 9 \quad \Rightarrow \quad V_2^{\cancel{max}} = 9V$$

$$-15 = \frac{4}{1.5} \times V_2 - 9 \quad \Rightarrow \quad V_2 = -2.25V$$

$$\Rightarrow \quad V_2^{max} = 9V$$

	Question 18-19 [4]		Vraag 18-19 [4]
18	If $v_1 = 200\ mV$ and $v_2 = 300\ mV$, calculate v_3.	**18**	Indien $v_1 = 200\ mV$ en $v_2 = 300\ mV$, bepaal v_3.
19	If $v_3 = 2\ V$, calculate v_o.	**19**	Indien $v_3 = 2\ V$, bepaal v_o.

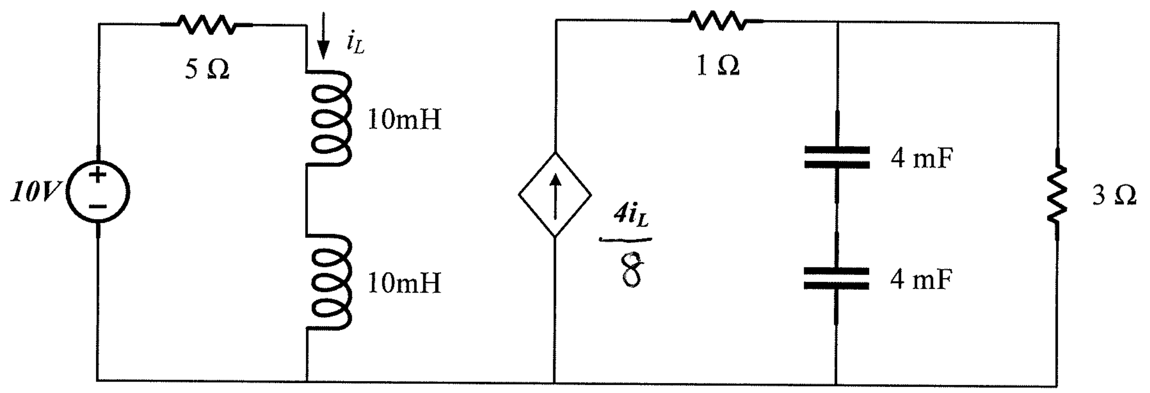


$$V_3 = -(0.2 \times 8 + 0.3 \times 2) = -2.2\ V$$

$$V_o = -2 \times \frac{12}{5} = -4.8\ V$$

Question 20-21 [4]	Vraag 20-21 [4]
If the circuit is in steady-state (DC conditions): 1) Determine total energy E_L stored in the inductors. 2) Determine total energy E_C stored in the capacitors.	Indien die kringbaan in gestadigde toestand is (GS toestande): 1) Bepaal totale energie E_L gestoor in die induktors. 2) Bepaal totale energie E_C gestoor in die kapasitors.

20	E_L	**20**	E_L
21	E_C	**21**	E_C



$i_L = 2A$, $V_C = 24V$

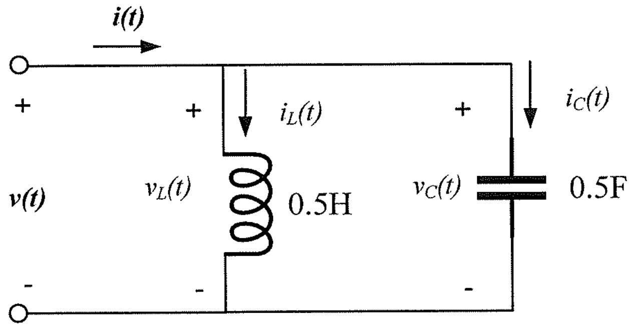
$L = 20mH$.

$C = 2mF$

$\Rightarrow E_L = \frac{1}{2} L i^2 = \frac{1}{2} \times 20 \times 10^{-3} \times 4 = 40mJ$

$E_C = \frac{1}{2} C V^2 = \frac{1}{2} \times 2 \times 10^{-3} \times 24^2 = 576\ mJ$

Question 22-23 [2]		**Vraag 22-23 [2]**	
If $v(t) = e^{-t}$ V, with $i_C(0) = 0\ A$ and $i_L(0) = 2\ A$, determine *i(t)*: $$i(t) = ae^{-t} + b$$ Give the coefficients *a* and *b*.		Indien $v(t) = e^{-t}$ V, met $i_C(0) = 0\ A$ en $i_L(0) = 2\ A$, bepaal *i(t)*: $$i(t) = ae^{-t} + b$$ Gee die koëffisiënte *a* en *b*.	
22	*a*	**22**	*a*
23	*b*	**23**	*b*



$$i_c(t) = C\frac{dv}{dt} = 0.5\,(e^{-t})' = -0.5\,e^{-t}$$

$$i_L(t) = \frac{1}{L}\int_0^t e^{-t}\,dt + 2$$

$$= 2\left[-e^{-t}\right]_0^t + 2$$

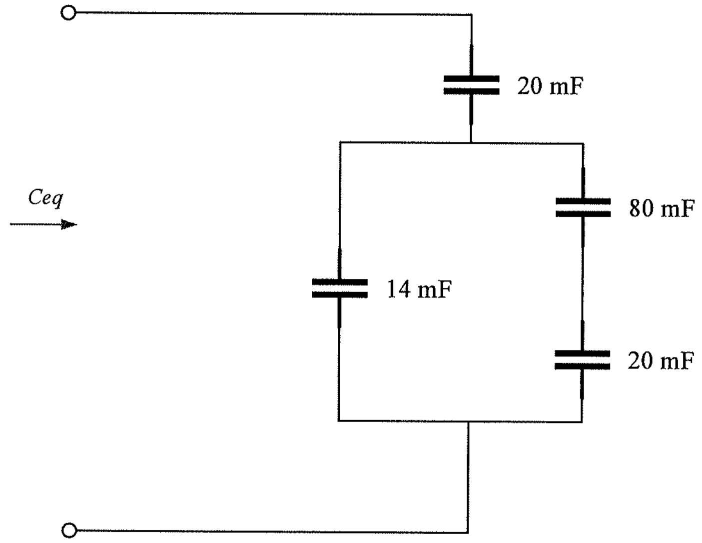
$$= 2(-e^{-t}+1) + 2 = -2e^{-t} + 4$$

$$i(t) = i_C(t) + i_L(t) = -2.5e^{-t} + 4$$

$$\Rightarrow a = -2.5$$

$$b = 4$$

Question 24 [2]	Vraag 24 [2]
Find the equivalent capacitance C_{eq} at terminals a-b.	Die ekwivalente kapasitansie C_{eq} is 12 mF. Bepaal die waarde van C.
24 C_{eq}	**26** C_{eq}



$$C_{eq} = \left(14 + \frac{80 \times 20}{80 + 20}\right) \| 20$$

$$= 30 \| 20$$

$$= 12\ mF$$

Question 25-28 [4]	Vraag 25-28 [4]
a) Simplify the following expression of $\bar{V}$ and write the answers in **polar form** $r\angle\theta°$. $$\bar{V} = \frac{\sqrt{50\angle 60°}}{(-5+j5)^*}$$	a) Vereenvoudig die volgende uitdrukking van $\bar{V}$ en skryf die antwoord in **reghoekige formaat** $a+jb$. $$\bar{V} = \frac{\sqrt{50\angle 60°}}{(-5+j5)^*}$$
b) Determine $\bar{V}_2$ in **rectangular form** $c+jd$. $$\begin{bmatrix} 3+j4 & -j2 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0° \\ 0 \end{bmatrix}$$	b) Bepaal $\bar{V}_2$ in **reghoekige formaat** $c+jd$. $$\begin{bmatrix} 3+j4 & -j2 \\ 2 & 5 \end{bmatrix}\begin{bmatrix} \bar{V}_1 \\ \bar{V}_2 \end{bmatrix} = \begin{bmatrix} 10\angle 0° \\ 0 \end{bmatrix}$$

25	r 0.48 − j0.64	**25**	r
26	θ	**26**	θ
27	c	**27**	c
28	d	**28**	d

$$\bar{V} = \frac{5\sqrt{2}\angle 30°}{5\sqrt{2}\angle -135°} = 1\angle 165°$$

$$\Rightarrow r = 1$$

$$\theta = 165°$$

$$\bar{V}_2 = \frac{\begin{vmatrix} 3+j4 & 10\angle 0° \\ 2 & 0 \end{vmatrix}}{\begin{vmatrix} 3+j4 & -j2 \\ 2 & 0 \end{vmatrix}} = \frac{-20}{\cancel{15+j20}+j4} = -2.5+j2.5$$

4

0.48 − j0.64

Questions 29-30 [2]		Vraag 29-30 [2]	
Given two sinusoid signals v_1 and v_2, where $v_1 = 10\cos(15t - 30°)$, $v_2 = -8\sin(15t + 150°)$. Determine $v = v_1 + v_2$, and write the final answer in the form $A\cos(15t + \phi)$ where $A > 0$ and $-180° \leq \phi \leq 180°$.		Gegee, twee sinusiede seine v_1 en v_2: $v_1 = 10\cos(15t - 30°)$, $v_2 = -8\sin(15t + 150°)$. Bepaal $v = v_1 + v_2$, en skryf die finale antwoord in die vorm $A\cos(15t + \phi)$ met $A > 0$ en $-180° \leq \phi \leq 180°$.	
29	A	**29**	A
30	ϕ	**30**	ϕ

$\bar{V}_1 = 10\angle -30°$

$\bar{U}_2 = 8\angle -120°$

$U = 12.806\angle -68.660°$

$A = 12.806$

$\phi = -68.660°$

PRACTICE PAGE / OEFENBLAD

Assessment ID	2	0	1	7	E	B	N	1	2	2	S	2		

Keep 5 significant numbers for your answers, add unit where applicable!

	Answer	Unit
01	$i_{o(15A)}=$ 1	A
02	$V_s=$ −60	V
03	$k_1=$ 40	*No unit required*
04	$k_2=$ 2	
05	$R=$ 8	Ω
06	$V_s=$ 80	V
07	$V_{Th}=$ 36	V
08	$R_{Th}=$ 20	Ω
09	$I_N=$ 72	A
10	$R_N=$ 2.2222	Ω
11	$V_{Th}=$ 40	V
12	$P_{max}=$ 40	W
13	$v_o=$ 12	V
14	$v_o=$ 2.4	V
15	$i_o=$ 0.2	mA
16	$v_o=$ 7	V
17	$v_2^{max}=$ 9	V
18	$v_3=$ 2.2	V
19	$v_o=$ −4.8	V
20	$E_L=$ 40	mJ
21	$E_C=$ 576	mJ
22	$a=$ −2.5	*No unit required*
23	$b=$ 4	
24	$C_{eq}=$ 12	mF
25	$\cancel{r}=$ 1	*No unit required*
26	$\theta=$ 165°	

27	$c =$ −2.5	
28	$d =$ 2.5	
29	$A =$ 12.806	
30	$\phi =$ −68.660°	